CASE 3 Intentional failure beam (UNSAFE detection) according to ACI 318-14

b x h = 25 x 50, cover = 3 cm, f'c = 240 ksc, fy = 4000 ksc, fyt = 2400 ksc, stirrup RB9@250

| Section | Top rebar | Bottom rebar | Side rebar | Mu (kg-m) | Vu (kg) | Tu (kg-m) | Pu (kg) |
|---|---|---|---|---|---|---|---|
| INITIAL | 5-DB25 | 2-DB12 | - | −6,000 | 8,000 | 0 | 0 |
| MID | 2-DB12 | 2-DB12 | - | 3,000 | 3,000 | 0 | 35,000 |
| END | Layer 1: 3-DB20, Layer 2: 2-DB20 (clear 2.5 cm) | 2-DB12 | - | −32,000 | 40,000 | 0 | 0 |

Test setup: Every section is intentionally non-compliant to verify that the program flags unsafe designs with the correct reasons. INITIAL: 5-DB25 does not fit within the net width (spacing error), the net tensile strain is below 0.004 (ductility failure), and the stirrup spacing (250 mm) exceeds $s_{max}$. MID: $A_s$ is below $A_{s,min}$ even though the flexural strength is adequate, the axial compression ($P_u$ = +35,000 kg) exceeds the beam applicability limit of $0.10f'_c \cdot Ag$, and the stirrup spacing exceeds $s_{max}$. END: Mu exceeds $\varphi M_n$ (the moment is also in the doubly reinforced branch of the design function), and Vu exceeds the shear capacity of the section. Only the failing checks are verified by hand calculation.1. Design Parameters and material properties

1.1. Cross-section geometry

Beam width (b) = 25 cm

Total depth (h) = 50 cm

Gross area ($A_g$) = b x h = 25 x 50 = 1,250 $cm^2$

Concrete cover = 3 cm

1.2. Material Specifications

Concrete compressive strength ($f_c'$) = 240 ksc

Main steel yield strength ($f_y$) = 4000 ksc (SD40)

Stirrup steel yield strength ($f_{yt}$) = 2400 ksc

Elasticity of steel = 2.04 x $10^6$ ksc

Stress block factor = 0.85 since ($f_c'$ = 240 < 280 ksc)

1.3. Strength reduction factors from table 21.2.1

Flexure = 0.65 to 0.90 (Determined by εt per Table 21.2.1)

Shear = 0.75

Torsion = 0.75

2. Critical section initial (support negative moment)

2.1 Geometry and factored demand

$M_u$ = -6,000 kg-m = -6.0 t-m

$V_u$ = 8,000 kg = 8.0 tons

$T_u$ = 0 kg-m = 0 t-m

$P_u$ = 0 kg = 0 tons

Top 5-DB25 ($A_s$ = 5 x 4.909 = 24.54 $cm^2$)

Bottom 2-DB12 ($A'_s$ = 2 x 1.131 = 2.26 $cm^2$)

Effective depth (d) of tension steel

$$y_1 = c_{cov} + d_{st} + \frac{d_b}{2} = 3.00 + 0.9 + \frac{2.50}{2} = 5.15 \text{ cm}$$

$$d = h - y_1 = 50.00 - 5.15 = 44.85 \text{ cm}$$

Compression steel depth from bottom face ($d_2$)

$$d' = c_{cov} + d_{st} + \frac{d_{b,12}}{2} = 3.00 + 0.9 + \frac{1.20}{2} = 4.50 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{clear} = \frac{b - 2c_{cov} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{25 - (2 \times 3.0) - (2 \times 0.9) - (5 \times 2.5)}{5 - 1} = 1.175 \text{ cm}$$

$$s_{clear,min} = \max(d_b, 2.5 \text{ cm}) = \max(2.5, 2.5) = 2.50 \text{ cm}$$

$s_{clear}(1.175 \text{ cm}) < s_{clear,min}(2.50 \text{ cm})$ FAIL

2.2. Flexural strength analysis

Force equilibrium and neutral axis depth

$$C_c + C'_s - T_s = P_u \; (where \; P_u = 0)$$

$$C_c + C'_s = T_s$$

Check yielding of tension steel $T_s$

Assume all layers of tension steel yield ($f_s = f_y$ = 4,000)

$$(0.85f'_c\beta_1 cb) + A'_s(f_y - 0.85f'_c) = T_s$$

$$T_s = A_s \cdot f_y = 24.54 \times 4{,}000 = 98{,}160 \text{ kg}$$

Check yielding of compression steel ($C'_s$)

Assume compression steel yields $f'_s = f_y = 4000 \; ksc$ accounting for displaced concrete area ($0.85f'_c$ = 204 ksc)

$$C'_s = A'_s \cdot (f'_s - 0.85f'_c) = 2.262 \times (4000 - 204) = 8{,}578.96 \text{ kg}$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 240 \times 0.85 \times c \times 25.0 = 4{,}335.0 \cdot c$$

Substitute values for c

$$4{,}335.0 \cdot c + 8{,}578.96 = 98{,}160 \text{ kg}$$

$$4{,}335.0 \cdot c = 98{,}160.0 - 8{,}578.96 = 89{,}581.04$$

$$c = \frac{89{,}581.04}{4{,}335.0} = 20.66 \text{ cm}$$

$$\epsilon'_s = 0.003 \cdot \left(\frac{c - d'}{c}\right) = 0.003 \cdot \left(\frac{20.66 - 4.50}{20.66}\right) = 0.00235$$

$$\epsilon_y = \frac{f_y}{E_s} = \frac{4000}{2{,}040{,}000} = 0.00196$$

Since $\epsilon'_s$ $(0.00235) \geq \epsilon_y$ $(0.00196)$

$$f'_s = f_y = 4{,}000 \text{ ksc} \quad \text{(YIELDED CONFIRMED)}$$

Check the depth of the compression block (a) compared to (d')

$$a = \beta_1 \cdot c = 0.85 \times 20.66 = 17.56 \text{ cm}$$

$$a \text{ (17.56 cm)} > d' \text{ (4.50 cm)}$$

Tension Strain Check ($\epsilon_t$) and Strength Reduction Factor (φ)

$$\epsilon_t = 0.003 \cdot \left(\frac{d_t - c}{c}\right) = 0.003 \cdot \left(\frac{44.85 - 20.66}{20.66}\right) = 0.00351$$

Code Requirement: $\epsilon_t \geq 0.004$ from ACI $318 - 14$

$\epsilon_t$ (0.00351) < 0.00400 FAIL (DUCTILITY FAILURE – OVER REINFORCED)

$$\phi = 0.65 + 0.25 \cdot \left(\frac{\epsilon_t - \epsilon_y}{0.00500 - \epsilon_y}\right) = 0.65 + 0.25 \cdot \left(\frac{0.00351 - 0.00196}{0.00500 - 0.00196}\right) = 0.78$$

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{240}}{4000}\right) \cdot 25 \cdot 44.85 = 3.92 \text{ cm}^2$$

Check with actual reinforcement (5-DB25 = 24.54 $cm^2$):

$A_{s,prov}(24.54\ cm^2) \geq A_{s,min}(3.92\ cm^2)$ PASS

Moment from concrete compression ($M_{Cc}$)

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 240 \times 0.85 \times 20.66 \times 25 = 89{,}561.1 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 25 - \frac{17.56}{2} = 16.22 \text{ cm}$$

$$M_{Cc} = 89{,}561.1 \times 16.22 = 1{,}452{,}681.04 \text{ kg-cm}$$

Moment from compression steel reinforcement ($M_{Cs}$)

$$C'_s = 8{,}578.96 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - d' = 25.00 - 4.50 = 20.50 \text{ cm}$$

$$M_{Cs} = 8{,}578.96 \times 20.50 = 175{,}868.68 \text{ kg-cm}$$

Moment from tension steel reinforcement ($M_{Ts}$)

$$T_s = 98{,}160 \text{ kg}$$

$$\text{Arm} = d - \frac{h}{2} = 44.85 - 25 = 19.85 \text{ cm}$$

$$M_{Ts} = 98{,}160 \times 19.85 = 1{,}948{,}476 \text{ kg-cm}$$

Total nominal flexural strength ($M_n$) and design flexural strength ($\phi M_n$)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 1{,}452{,}681.04 + 175{,}868.68 + 1{,}948{,}476 = 3{,}577{,}025.72 \text{ kg-cm} = 35.77 \text{ t-m}$$

$$\phi M_n = 0.78 \times 35.77 = 27.90 \text{ t-m}$$

Design moment capacity ($\phi M_n$)

$\phi M_n$ (27.90 t-m) $\geq |M_u|$ (6 t-m) PASS (Only flexurual demand)

2.3. Shear capacity analysis

Concrete shear resistance from ACI 318-14

$$V_c = 0.53\sqrt{f'_c} \cdot b \cdot d = 0.53 \times \sqrt{240} \times 25 \times 44.85 = 9{,}206 \text{ kg} = 9.21 \text{ tons}$$

Provided tranverse stirrup shear strength ($V_s$ - 2 Legs RB9@ 25 cm)

$$A_v = 2 \times \frac{\pi(0.90)^2}{4} = 1.27 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.27 \times 2400 \times 44.85}{25} = 5{,}468 \text{ kg} = 5.47 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{min}} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.036 \text{ cm}^2/\text{cm}$$

Check with actual stirrup reinforcement (2-legged RB9@ 25 cm, $A_v = 1.27$ cm²)

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.27}{25} = 0.051 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} (0.051 \text{ cm}^2/\text{cm}) \geq \left(\frac{A_v}{s}\right)_{\text{min}} (0.036 \text{ cm}^2/\text{cm}) \text{ PASS}$$

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\max} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{240} \times 25 \times 44.85 = 36{,}477.69 \text{ kg} = 36.48 \text{ tons}$$

$$V_{s,\max} (36.48 \text{ tons}) \geq V_s (5.47 \text{ tons}) \text{ PASS}$$

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{240} \times 25 \times 44.85 = 19{,}107.36 \text{ kg} = 19.11 \text{ tons}$$

$$V_s (5.47 \text{ tons}) \leq \text{Threshold limit } (19.11 \text{ tons})$$

$$s_{\max} = \min\left(\frac{d}{2}, 60.0 \text{ cm}\right) = \min\left(\frac{44.85}{2}, 60.0\right) = 22.43 \text{ cm}$$

$$s_{\max} (22.40 \text{ cm}) \leq s_{\text{prov}} (25.00 \text{ cm}) \text{ NOTPASS (spacing exceeded)}$$

Total design shear strength ($\phi V_n$)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.75 \times (9.21 + 5.47) = 11.01 \text{ tons}$$

$$\phi V_n (11.01 \text{ tons}) \geq V_u (8 \text{ tons}) \text{ PASS (shear demand)}$$

2.4. Torsion assessment

Threshold torsion calculation ($T_{th}$)

$$A_{cp} = b \cdot h = 25 \times 50 = 1{,}250.00 \text{ cm}^2, \quad p_{cp} = 2(b + h) = 150 \text{ cm}$$

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right) = 0.75 \times 0.27 \times 1.0 \times \sqrt{240} \times \left(\frac{1250^2}{150}\right) = 32{,}678.30 \text{ kg-cm}$$

Torsion Design

$$T_{th} (0.33 \text{ t-m}) \geq T_u (0 \text{ t-m}) \text{ NEGLECT TORSION}$$

## 3. Critical section MID (span positive moment)

3.1 Geometry and factored demand

$M_u$ = 3,000 kg-m = 3 t-m

$V_u$ = 3,000 kg = 3 tons

$T_u$ = 0 kg-m = 0 t-m

$P_u$ = 35,000 kg = 35 tons

Top 2-DB12 ($A'_s$ = 2 x 1.131 = 2.262 $cm^2$)

Bottom 2-DB12 ($A_s$ = 2 x 1.131 = 2.262 $cm^2$)

Beam Applicability Limit Check (ACI 318-14 Section 9.3.3.1)

The factored axial compressive force on a flexural member shall not exceed ($0.1f'_cA_g$)

$$P_{\text{limit}} = 0.10f'_cA_g = 0.10 \times 240 \times 1{,}250 = 30{,}000 \text{ kg} = 30 \text{ tons}$$

$P_{limit}(30{,}000 \text{ kg}) \leq P_u(35{,}000 \text{ kg})$ NOTPASS (exceeds beam limit)

Effective depth (d) of tension steel (bottom)

$$y_1 = c_{\text{cov}} + d_{st} + \frac{d_b}{2} = 3 + 0.9 + \frac{1.20}{2} = 4.50 \text{ cm}$$

$$d = h - y_1 = 50 - 4.5 = 45.5 \text{ cm}$$

Compression steel depth from top face ($d_2$)

$$d' = c_{\text{cov}} + d_{st} + \frac{d_{b,12}}{2} = 3 + 0.9 + \frac{1.20}{2} = 4.50 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{\text{clear}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{25 - (2 \times 3.0) - (2 \times 0.9) - (2 \times 1.2)}{2 - 1} = 14.80 \text{ cm}$$

$$s_{\text{clear,min}} = \max(d_b,\ 2.5 \text{ cm}) = \max(1.2,\ 2.5) = 2.50 \text{ cm}$$

$s_{\text{clear}}(14.80 \text{ cm}) \geq s_{\text{clear,min}}(2.50 \text{ cm})$ PASS

3.2. Flexural strength analysis

Force equilibrium and neutral axis depth

$$C_c + C'_s - T_s = P_u$$

$$C_c + C'_s = P_u + T_s$$

Check yielding of tension steel $T_s$

Assume tension steel yields ($f_s = f_y$ = 4,000)

$$T_s = A_s \cdot f_y = 2.26 \times 4000 = 9{,}040 \text{ kg}$$

$$T_s + P_u = 9{,}040 + 35{,}000 = 44{,}040 \text{ kg}$$

Check yielding of compression steel ($C'_s$)

The stress in compression steel is $f'_s = \min(f_y,\ E_s\epsilon'_s)$ accounting for the displaced concrete area ($0.85f'_c = 204$ ksc)

Since neutral axis c is near d', compression steel remains elastic ($f'_s < f_y$).

$$f'_s = E_s \cdot \epsilon'_s = 2.04 \times 10^6 \times 0.003 \cdot \frac{c - d'}{c} = 6{,}120 \cdot \frac{c - 4.5}{c}$$

$$C'_s = A'_s(f'_s - 0.85f'_c) = 2.262\left(6{,}120 \cdot \frac{c - 4.5}{c} - 204\right) = 13{,}843.44 \cdot \frac{c - 4.5}{c} - 461.45$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 240 \times 0.85 \times c \times 25.0 = 4{,}335.0 \cdot c$$

Solving equilibrium quadratic equation for neutral axis c:

$$4{,}335 \cdot c + 13{,}843.44\left(\frac{c - 4.50}{c}\right) - 461.45 = 44{,}040$$

$$4{,}335c^2 - 30{,}658c - 62{,}295 = 0$$

$$c = 8.72 \text{ cm}$$

$$\epsilon'_s = 0.003 \cdot \left(\frac{c - d'}{c}\right) = 0.003 \cdot \frac{8.72 - 4.5}{8.72} = 0.00145 < \epsilon_y = 0.00196$$

(NOT YIELD CONFIRMED)

$$f'_s = 2.04 \times 10^6 \times 0.00145 = 2{,}958 \text{ ksc} < f_y = 4{,}000 \text{ ksc}$$

Check the depth of the compression block (a) compared to (d')

$$a = \beta_1 \cdot c = 0.85 \times 8.72 = 7.41 \text{ cm}$$

$$a\ (7.41 \text{ cm}) > d'\ (4.50 \text{ cm})$$

Tension Strain Check ($\epsilon_t$) and Strength Reduction Factor (φ)

$$\epsilon_{t,\text{outer}} = 0.003 \cdot \left(\frac{d - c}{c}\right) = 0.003 \cdot \left(\frac{45.5 - 8.72}{8.72}\right) = 0.01265$$

$$\epsilon_{t,\text{outer}} = 0.01265 \geq 0.005$$

$$\phi = 0.900 \text{ (Tension} - \text{Controlled)}$$

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{240}}{4000}\right) \cdot 25 \cdot 45.5 = 3.98 \text{ cm}^2$$

Check with actual reinforcement (2-DB12 = 2.26 cm$^2$):

$A_{s,prov}(2.26\ cm^2) \leq A_{s,min}(3.98\ cm^2)$ NOTPASS (insufficient Minimum Reinforcement)

Moment from concrete compression ($M_{Cc}$)

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 240 \times 0.85 \times 8.72 \times 25.0 = 37{,}801.2 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 25 - \frac{7.41}{2} = 21.30 \text{ cm}$$

$$M_{Cc} = 37{,}801 \times 21.30 = 805{,}161 \text{ kg-cm}$$

Moment from compression steel reinforcement ($M_{Cs}$)

$$C'_s = A'_s \cdot (f'_s - 0.85f'_c) = 2.262 \times (2{,}958 - 204) = 6{,}224 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - d' = 25 - 4.5 = 20.5 \text{ cm}$$

$$M_{Cs} = 6{,}224 \times 20.5 = 127{,}592 \text{ kg-cm}$$

Moment from tension steel reinforcement ($M_{Ts}$)

$$T_s = 9{,}040 \text{ kg}$$

$$\text{Arm} = d - \frac{h}{2} = 45.5 - 25 = 20.5 \text{ cm}$$

$$M_{Ts} = 9{,}040 \times 20.5 = 185{,}320 \text{ kg-cm}$$

Total nominal flexural strength ($M_n$) and design flexural strength ($\phi M_n$)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 805{,}161 + 127{,}592 + 185{,}320 = 1{,}118{,}073 \text{ kg-cm} = 11.18 \text{ t-m}$$

$$\phi M_n = 0.90 \times 11.18 = 10.06 \text{ t-m}$$

Design moment capacity ($\phi M_n$)

$\phi M_n$ (10.06 t-m) $\geq |M_u|$ (3 t-m) PASS (flexure demand)

3.3. Shear capacity analysis

Concrete shear resistance from ACI 318-14 Table 22.5.6.1

$$V_c = 0.53 \cdot \left(1 + \frac{P_u}{140 \cdot A_g}\right) \cdot \sqrt{f'_c} \cdot b \cdot d$$

$$\left(1 + \frac{P_u}{140 \cdot A_g}\right) = 1 + \frac{35{,}000}{140 \times 1{,}250} = 1 + 0.20 = 1.20$$

$$V_c = 0.53 \times 1.20 \times \sqrt{240} \times 25 \times 45.5 = 11{,}207 \text{ kg} = 11.21 \text{ tons}$$

$$\phi V_c = 0.75 \times 11.21 = 8.41 \text{ tons}$$

Provided Tranverse stirrup shear strength ($V_s$ - 2 Legs RB9@ 25 cm)

$$A_v = 2 \times \frac{\pi(0.9)^2}{4} = 1.27 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.27 \times 2400 \times 45.5}{25} = 5{,}547.36 \text{ kg} = 5.55 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.27}{25} = 0.051 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\min} = \max\left(\frac{0.2\sqrt{f'_c}\cdot b}{f_{yt}}, \frac{3.5\cdot b}{f_{yt}}\right) = 0.036\ \text{cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} (0.051\ \text{cm}^2/\text{cm}) \geq \left(\frac{A_v}{s}\right)_{\min} (0.036\ \text{cm}^2/\text{cm})\ \text{PASS}$$

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\max} = 2.1\sqrt{f'_c}\cdot b\cdot d = 2.1\ \times\sqrt{240}\times 25\times 45.50 = 37{,}006.36\ \text{kg} = 37\ \text{tons}$$

$$V_{s,\max}\ (37\ \text{tons}) \geq V_s\ (5.55\ \text{tons})\ \text{PASS}$$

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c}\cdot b\cdot d = 1.10\times\sqrt{240}\times 25\times 45.5 = 19{,}384.28\ \text{kg}$$

$$V_s\ (5.55\ \text{tons}) \leq \text{Threshold limit}\ (19.38\ \text{tons})$$

$$s_{\max} = \min\left(\frac{d}{2}, 60.0\ \text{cm}\right) = \min\left(\frac{45.5}{2}, 60\right) = 22.75\ \text{cm}$$

$$s_{\max}\ (22.75\ \text{cm}) \leq s_{\text{prov}}\ (25\ \text{cm})\ \text{NOTPASS (spacing exceeded)}$$

Total design shear strength ($\phi V_n$)

$$\phi V_n = \phi_{\text{shear}}\cdot(V_c + V_s) = 0.75\times(11.21 + 5.55) = 12.57\ \text{tons}$$

$$\phi V_n\ (12.57\ \text{tons}) \geq V_u\ (3\ \text{tons})\ \text{PASS (shear demand)}$$

3.4. Torsion assessment

Threshold torsion calculation ($T_{th}$) (ACI 318-14 Sec 22.7.4.1)

$$A_{cp} = b\cdot h = 25\times 50 = 1{,}250\ \text{cm}^2, \quad p_{cp} = 2(b+h) = 150\ \text{cm}$$

$$T_{th} = \phi_{\text{torsion}}\cdot 0.27\lambda\sqrt{f'_c}\cdot\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1+\frac{P_u}{0.33A_g\sqrt{f'_c}}}$$

$$T_{th} = 0.75\times 0.27\times 1.0\times\sqrt{240}\times\left(\frac{1250^2}{150}\right)\times\sqrt{1+\frac{35000}{0.33\times 1250\times\sqrt{240}}} = 35{,}765\ \text{kg-cm}$$

Torsion Design

$$T_{th}\ (0.33\ \text{t-m}) \geq T_u\ (0.0\ \text{t-m})\ \text{NEGLECT TORSION}$$

## 4. Critical section END (support negative moment)

### 4.1. Geometry and factored demand

$M_u$ = -32,000 kg-m = -32 t-m

$V_u$ = 40,000 kg = 40 tons

$T_u$ = 0 kg-m = 0 t-m

$P_u$ = 0 kg = 0 tons

Top 3-DB20 ($A_s$ = 3 x 3.142 = 9.43 $cm^2$), Layer 2 2-DB20 ($A_s$ = 2 x 3.142 = 6.28 $cm^2$), Total = 15.71 $cm^2$ distance = 2.5 cm

Bottom 2-DB12 ($A'_s$ = 2 x 1.131 = 2.26 $cm^2$)

Effective depth (d) of tension steel (top)

$$y_1 = c_{\text{cov}} + d_{st} + \frac{d_b}{2} = 3 + 0.9 + \frac{2.0}{2} = 4.9 \text{ cm}$$

$$d_1 = h - y_1 = 50 - 4.9 = 45.1 \text{ cm}$$

$$y_2 = y_1 + d_b + s_{\text{clear,vert}} = 4.9 + 2.0 + 2.5 = 9.4 \text{ cm}$$

$$\bar{y} = \frac{(3 \times 4.9) + (2 \times 9.4)}{5} = \frac{14.7 + 18.8}{5} = 6.7 \text{ cm}$$

$$d = h - \bar{y} = 50 - 6.7 = 43.3 \text{ cm}$$

Compression steel depth from bottom face (d')

$$d' = c_{\text{cov}} + d_{st} + \frac{d_{b,12}}{2} = 3 + 0.9 + \frac{1.2}{2} = 4.5 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{\text{clear}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{25 - (2 \times 3.0) - (2 \times 0.9) - (3 \times 2.0)}{3 - 1} = 5.60 \text{ cm}$$

$$s_{\text{clear,min}} = \max(d_b,\ 2.5 \text{ cm}) = \max(2.0,\ 2.5) = 2.50 \text{ cm}$$

$s_{\text{clear}}$(5.60 cm) ≥ $s_{\text{clear,min}}$(2.50 cm) PASS

4.2. Flexural strength analysis

Force equilibrium and neutral axis depth

$$C_c + C'_s - T_s = P_u \ (where\ P_u\ =\ 0)$$

$$C_c + C'_s = T_s$$

Check yielding of tension steel $T_s$

Assume all layers of tension steel yield ($f_{s1}$ = $f_{s2}$ = $f_y$ = 4,000)

$$T_s = A_s \cdot f_y = 15.71 \times 4{,}000 = 62{,}840 \text{ kg}$$

Check yielding of compression steel ($C'_s$)

The stress in compression steel is $f'_s = \min(f_y, E_s\epsilon'_s)$ accounting for the displaced concrete area ($0.85f'_c = 204$ ksc)

Solving equilibrium quadratic equation for neutral axis c:

$$f'_s = E_s \cdot \epsilon'_s = 2.04 \times 10^6 \times 0.003 \cdot \frac{c - d'}{c} = 6{,}120 \cdot \frac{c - 4.5}{c}$$

$$C'_s = A'_s(f'_s - 0.85f'_c) = 2.26\left(6{,}120 \cdot \frac{c - 4.5}{c} - 204\right) = 13{,}831 \cdot \frac{c - 4.5}{c} - 461$$

$$C_c = 0.85f'_c\beta_1 cb = 0.85 \times 240 \times 0.85 \times c \times 25 = 4{,}335 \cdot c$$

$$4{,}335 \cdot c + 13{,}831 \cdot \frac{c - 4.5}{c} - 461 = 62{,}840$$

$$4{,}335 \cdot c^2 - 49{,}470c - 62{,}240 = 0$$

$$c = 12.55 \text{ cm}$$

$$\epsilon'_s = 0.003 \cdot \left(\frac{12.55 - 4.5}{12.55}\right) = 0.00192 < \epsilon_y(0.00196) \quad \text{(NOT YIELD)}$$

$$f'_s = 2.04 \times 10^6 \times 0.00192 = 3{,}917 \text{ ksc} < 4{,}000 \text{ ksc}$$

$$C'_s = A'_s \cdot (f'_s - 0.85f'_c) = 2.26 \times (3{,}917 - 204) = 8{,}391 \text{ kg}$$

$$C_c = 0.85f'_c\beta_1 cb = 4{,}335 \times 12.55 = 54{,}404 \text{ kg}$$

Check the depth of the compression block (a) compared to (d')

$$a = \beta_1 \cdot c = 0.85 \times 12.55 = 10.67 \text{ cm} > d' = 4.5 \text{ cm}$$

Tension Strain Check ($\epsilon_t$) and Strength Reduction Factor (φ)

$$\epsilon_t = 0.003 \cdot \frac{d_t - c}{c} = 0.003 \cdot \frac{45.1 - 12.55}{12.55} = 0.00778$$

$$\epsilon_{t,\text{outer}} = 0.00778 \geq 0.005$$

$$\phi = 0.900 \text{ (Tension} - \text{Controlled)}$$

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{240}}{4000}\right) \cdot 25 \cdot 43.3 = 3.79 \text{ cm}^2$$

$A_{s,prov}(15.71\ cm^2) \geq A_{s,min}(3.79\ cm^2)$ PASS

Moment from concrete compression ($M_{Cc}$)

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 25 - \frac{10.67}{2} = 19.67 \text{ cm}$$

$$M_{Cc} = 54{,}404 \times 19.67 = 1{,}070{,}127 \text{ kg-cm}$$

Moment from compression steel reinforcement ($M_{Cs}$)

$$\text{Arm} = \frac{h}{2} - d' = 25 - 4.5 = 20.5 \text{ cm}$$

$$M_{Cs} = 8{,}391 \times 20.5 = 172{,}016 \text{ kg-cm}$$

Moment from tension steel reinforcement ($M_{Ts}$ about the centroid axis h/2)

$$\text{Arm} = d - \frac{h}{2} = 43.3 - 25 = 18.3 \text{ cm}$$

$$M_{Ts} = 62{,}840 \times 18.3 = 1{,}149{,}972 \text{ kg-cm}$$

Total nominal flexural strength ($M_n$) and design flexural strength ($\phi M_n$)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 1{,}070{,}127 + 172{,}016 + 1{,}149{,}972 = 2{,}392{,}115 \text{ kg-cm} = 23.92 \text{ t-m}$$

$$\phi M_n = 0.90 \times 23.92 = 21.53 \text{ t-m}$$

Design moment capacity ($\phi M_n$)

$\phi M_n$ (21.53 t-m) $\leq |M_u|$ (32.00 t-m) NOTPASS (flexural failure)

4.3. Shear capacity analysis

Concrete shear resistance from ACI 318-14 Table 22.5.5.1

$$V_c = 0.53\sqrt{f'_c} \cdot b \cdot d = 0.53 \times \sqrt{240} \times 25 \times 43.3 = 8{,}889 \text{ kg} = 8.89 \text{ tons}$$

Provided Tranverse stirrup shear strength ($V_s$ - 2 Legs RB9@ 25 cm)

$$A_v = 2 \times \frac{\pi(0.9)^2}{4} = 1.27 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s} = \frac{1.27 \times 2{,}400 \times 43.3}{25} = 5{,}279 \text{ kg} = 5.28 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.27}{25} = 0.051 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\text{min}} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.036 \text{ cm}^2/\text{cm}$$

$\left(\frac{A_v}{s}\right)_{\text{prov}}$ $(0.051 \text{ cm}^2/\text{cm}) \geq \left(\frac{A_v}{s}\right)_{\text{min}}$ $(0.036 \text{ cm}^2/\text{cm})$ PASS

Total design shear strength

$$\phi V_n = \phi_{\text{shear}}(V_c + V_s) = 0.75 \times (8.89 + 5.28) = 10.63 \text{ tons}$$

$\phi V_n$ (10.63 tons) $\geq V_u$ (40.00 tons) NOTPASS (shear failure)

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\text{req}} = \frac{V_u}{\phi} - V_c = \frac{40{,}000}{0.75} - 8{,}889 = 53{,}333 - 8{,}889 = 44{,}444 \text{ kg} = 44.44 \text{ tons}$$

$$V_{s,\max} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{240} \times 25 \times 43.3 = 35{,}212 \text{ kg} = 35.21 \text{ tons}$$

$V_{s,\max}$ (35.21 tons) $\leq V_{s,req}$ (44.44 tons) NOTPASS (concrete crushing

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.1 \times \sqrt{240} \times 25 \times 43.3 = 18{,}445 \text{ kg} = 18.45 \text{ tons}$$

$$V_s \text{ (5.28 tons)} \leq \text{Threshold limit (18.45 tons)}$$

$$s_{\max} = \min\left(\frac{d}{2}, 60.0 \text{ cm}\right) = \min\left(\frac{43.30}{2}, 60\right) = 21.65 \text{ cm}$$

$s_{\max}$ (21.65 cm) $\leq s_{\text{prov}}$ (25 cm) NOTPASS (spacing exceeded)

4.4. Torsion assessment

$$A_{cp} = 25 \times 50 = 1{,}250 \text{ cm}^2,\ p_{cp} = 2(25 + 50) = 150 \text{ cm}$$

Threshold torsion calculation ($T_{th}$)

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right) = 0.75 \times 0.27 \times 1.0 \times \sqrt{240} \times \left(\frac{1250^2}{150}\right) = 32{,}679 \text{ kg-cm}$$

Torsion Design

$T_{th}$ (0.33 t-m) $\geq T_u$ (0 t-m) NEGLECT TORSION

| Section | Governing Action Demand | Design Capacity | D/C Ratio (UR) | ACI 318-14 Compliance Status |
|---|---|---|---|---|
| INITIAL (Support Negative) | $M_u = -6.0$ t-m | $\phi M_n = 27.79$ t-m | 0.22 | FAIL (UNSAFE: $\epsilon_t = 0.0035 < 0.004$ Ductility Failure, Rebar Clear Spacing $s_{\text{clear}} = 1.18$ cm $< 2.50$ cm) |
| | $V_u = 8.0$ tons | $\phi V_n = 11.01$ tons | 0.73 | FAIL (UNSAFE: Stirrup Spacing $s_{\text{prov}} = 25.0$ cm $> s_{\max} = 22.4$ cm) |
| | $T_u = 0$ t-m | $T_{th} = 0.33$ t-m | 0 | PASS (Neglect Torsion per $T_u \leq T_{\text{th}}$) |
| MID (Span Positive) | $M_u = 3.0$ t-m | $\phi M_n = 10.06$ t-m | 1.17 | FAIL (UNSAFE: Insufficient Minimum Reinforcement $A_{s,\text{prov}} = 2.26$ cm$^2$ $< A_{s,\min} = 3.98$ cm$^2$) |
| | $V_u = 3.0$ tons | $\phi V_n = 12.57$ tons | 0.3 | FAIL (UNSAFE: Stirrup Spacing $s_{\text{prov}} = 25.0$ cm $> s_{\max} = 22.8$ cm) |
| | $P_u = 35.0$ t-m | $P_{\text{limit}} = 30.0$ tons | 0.24 | FAIL (UNSAFE: Axial Compression Exceeds Beam Applicability Limit $0.10 f'_c A_g$, Must be designed as Beam-Column) |
| | $T_u = 0$ t-m | $T_{th} = 0.36$ t-m | 0 | PASS (Neglect Torsion per $T_u \leq T_{\text{th}}$) |
| END (Support Negative) | $M_u = -32.0$ t-m | $\phi M_n = 21.53$ t-m | 1.49 | FAIL (UNSAFE: Flexural Overstress, $\|M_u\| > \phi M_n$) |
| | $V_u = 40.0$ tons | $\phi V_n = 10.63$ tons | 3.76 | FAIL (UNSAFE: Severe Shear Failure $V_u \gg \phi V_n$, Concrete Web Crushing $V_{s,\text{req}} = 44.44$ t $> V_{s,\max} = 35.21$ t) |
| | $T_u = 0$ t-m | $T_{th} = 0.33$ t-m | 0 | PASS (Neglect Torsion per $T_u \leq T_{\text{th}}$) |